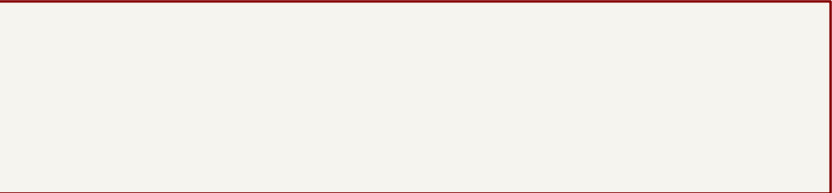


FarmMesh Node / Revision A0 – DRAFT

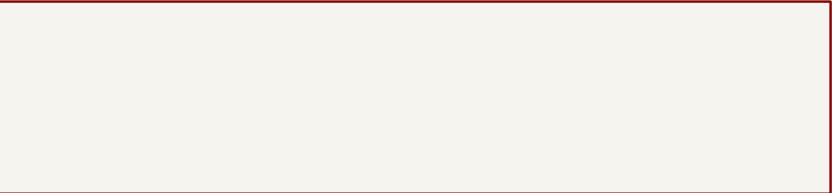
1 x RP2350B + 3 x ESP32–C5–WROOM–1U–N8R8
1S protected Li–ion input; 3.3V regulated rail
Native KiCad 10 schematic – engineering review only

1S battery input and 3.3V buck–boost



File: 01–power.kicad_sch

RP2350B aggregation, memory and debug



File: 02–rp2350.kicad_sch

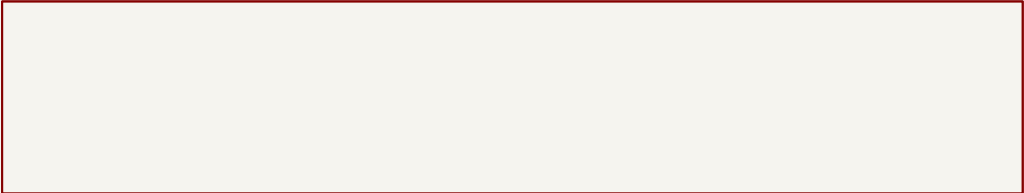
INTERFACE CONVENTION

TX / RX are named from the C5 perspective.
Both TX_CLK and RX_CLK originate at that C5.
RP drives only RX_D[0..3] and RX_VALID.
Global net labels make the electrical sheet connections.
No wireless protocol is selected by this schematic.

REVIEW BOUNDARIES

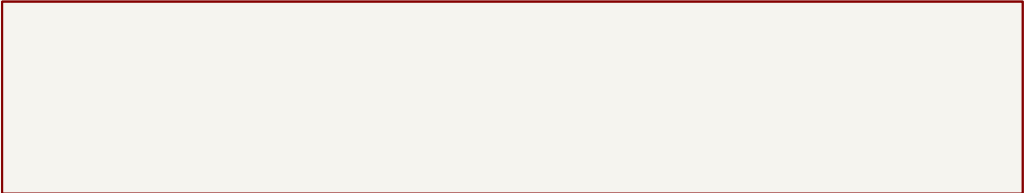
Battery: conventional 4.20V–max Li–ion, NOT LiFePO4 or HV Li–ion.
Normal loaded operation about 3.0–4.2V; cold start about 3.3V.
External protected pack required. No on–board charger or PV input.
PCB placement, RF isolation, converter stability and throughput unvalidated.
See DESIGN–NOTES.md and reports/erc.json before any layout or procurement.

C5–A module and independent link A



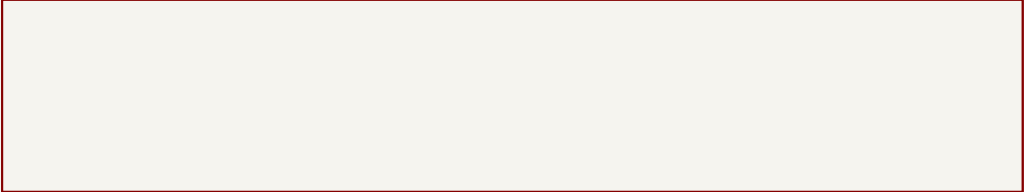
File: 03–radio–a.kicad_sch

C5–B module and independent link B



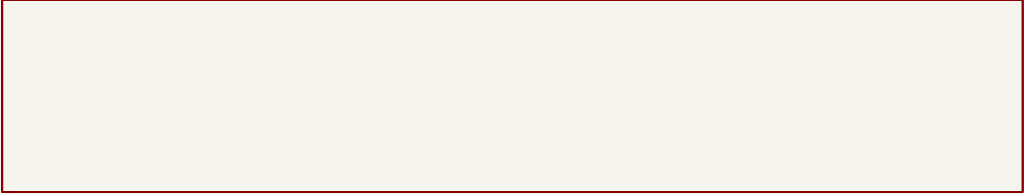
File: 04–radio–b.kicad_sch

C5–C module and independent link C



File: 05–radio–c.kicad_sch

RP USB boot / self–powered USB–C



File: 06–usb.kicad_sch

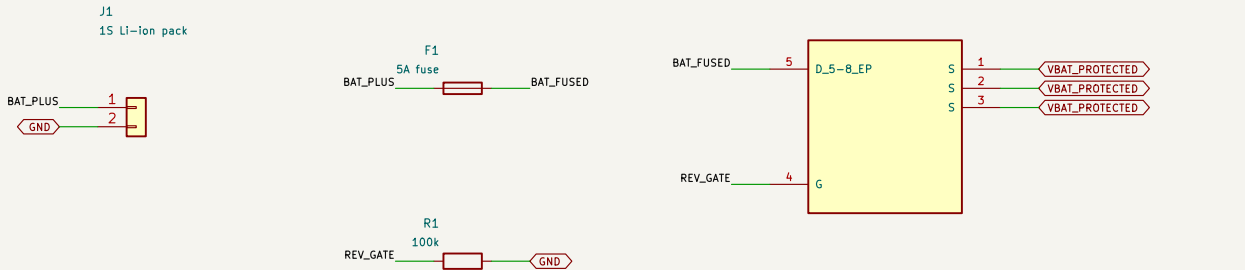
1S Li–ion battery input; no on–board charger
Engineering review only – NOT FOR FABRICATION

FarmMesh Node
Sheet: /
File: FarmMesh–Node.kicad_sch

Title: FarmMesh Node – schematic review index

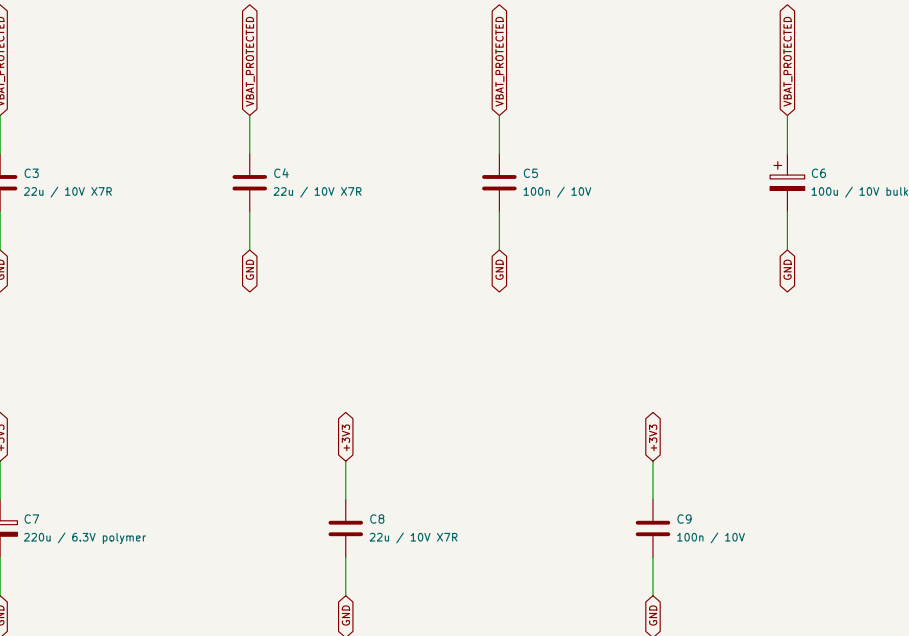
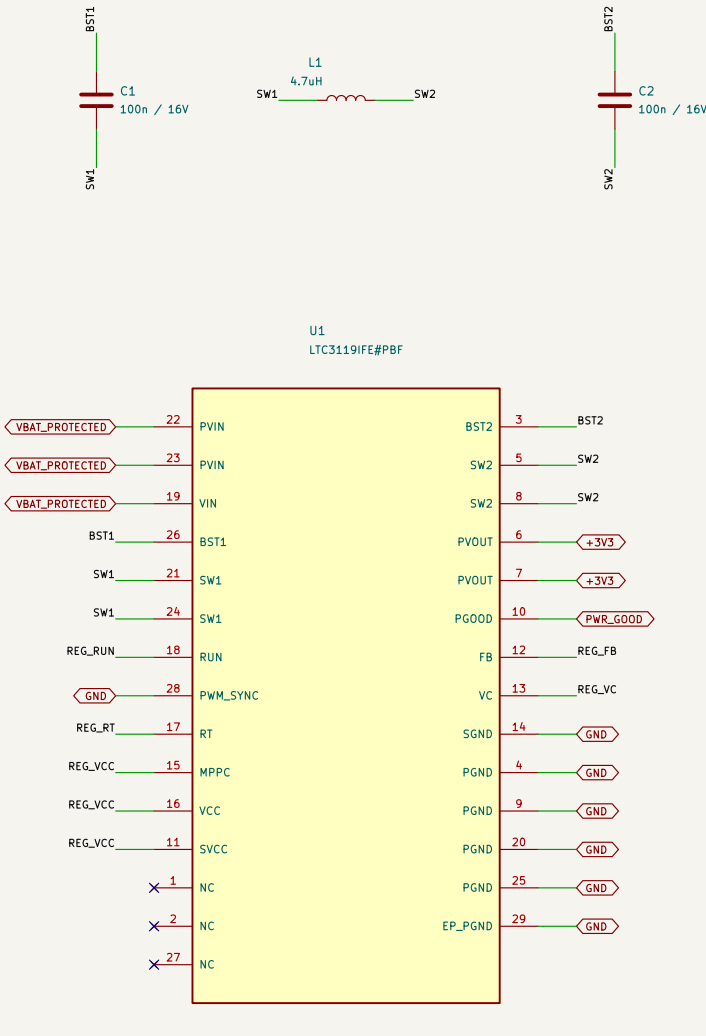
Size: A3	Date: 2026–09–23	Rev: A0 DRAFT
KiCad E.D.A. 10.0.5		Id: 1/7

BATTERY INPUT / REVERSE POLARITY

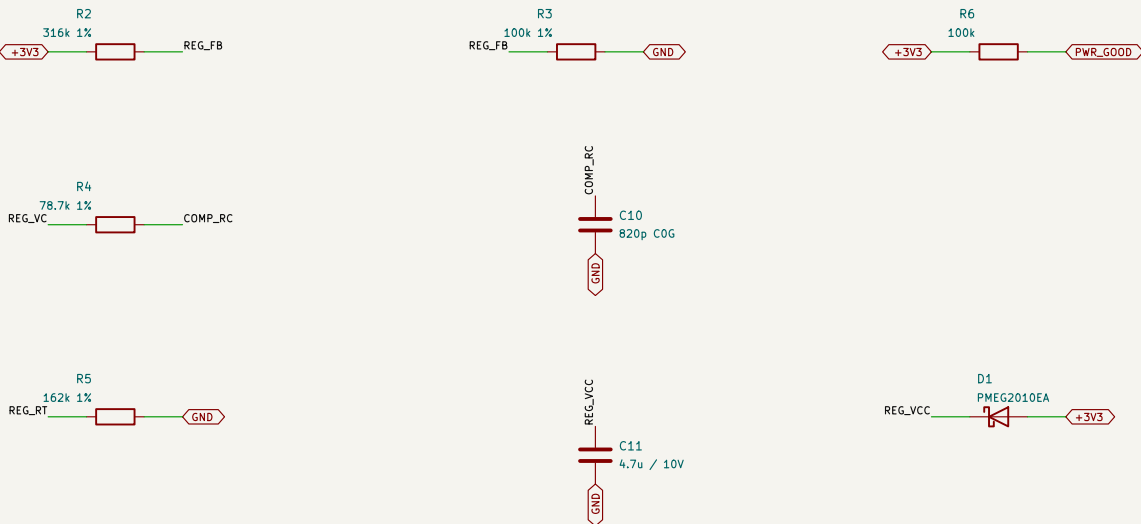


Q1 drain faces battery, source faces converter.
Body diode initially conducts toward load.
Protected pack / connector must support >=5A peak.
Fuse is not a substitute for cell protection.

3.3V BUCK-BOOST / LTC3119 TA06 BASIS

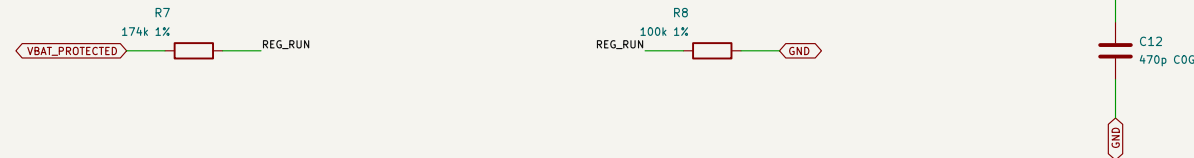


OUTPUT: 2.2A rail budget; 3A transient validation target.
3 x C5 supply capability >=0.6A each + RP and margin.
Not a guaranteed continuous 3A rating at every input/temperature.



500kHz; VOUT = 0.795 x (1 + 316k/100k) = 3.307V nominal.
MPPC tied to VCC (disabled). PWM/SYNC = GND (Burst allowed).
SGND and PGND meet at exposed-pad ground; honor TA06 layout.
D1: anode=3V3, cathode=VCC; do not short VCC to 3V3.

BATTERY UVLO / USER OFF



J2
OFF jumper

RUN nominal: on 3.30V / off 3.01V; tolerances/load sag apply.
External charger/PV is not connected in this draft.

1S Li-ion battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION
FarmMesh Node
Sheet: /1S battery input and 3.3V buck-boost/
File: 01-power.kicad_sch

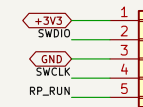
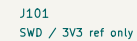
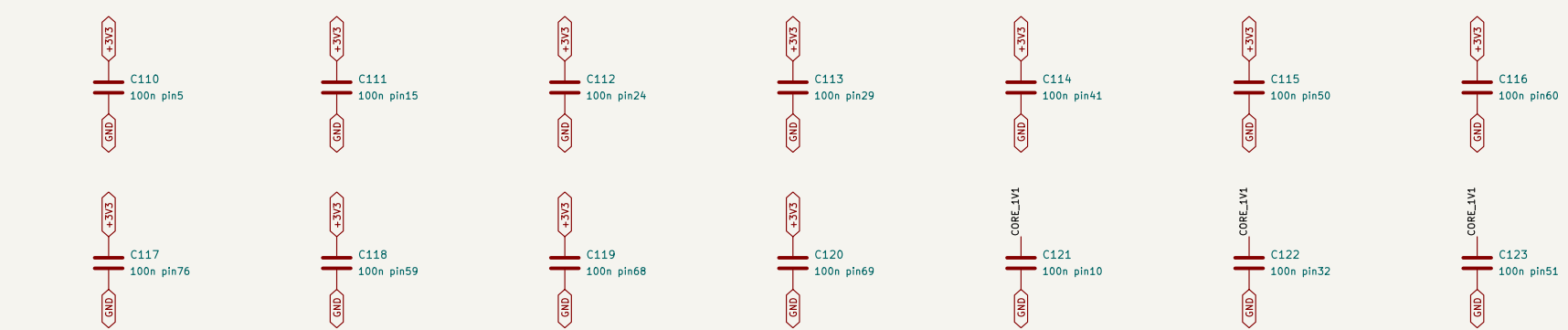
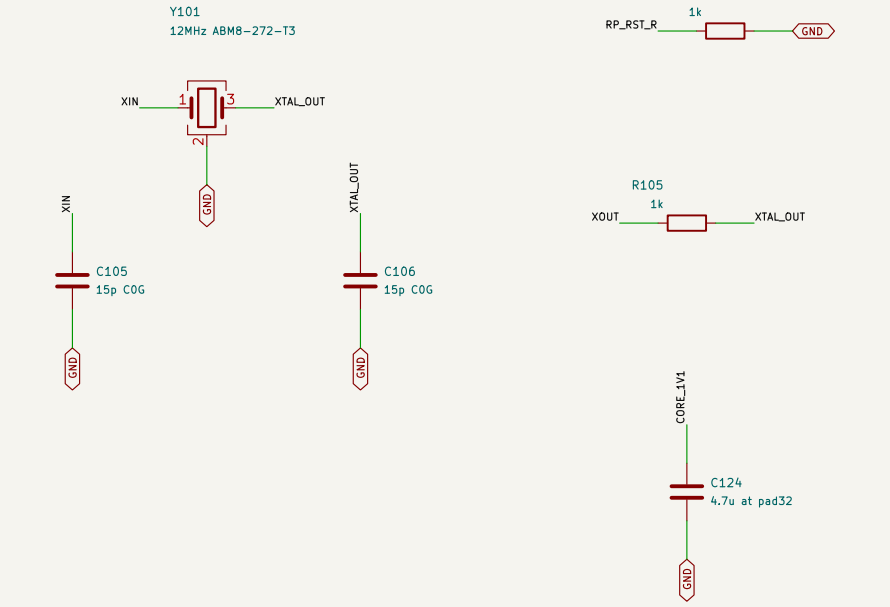
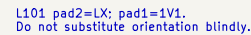
Title: 1S battery input and 3.3V buck-boost

Size: A3
KiCad E.D.A. 10.0.5

Date: 2026-09-23

Rev: A0 DRAFT
Id: 2/7

CORE BUCK – follow official R4-S1 placement and L orientation



PIO0 BASE0: GPIO0..11 = A
 PIO1 BASE0: GPIO12..23 = B
 PIO2 BASE16: GPIO24..35 = C
 36 = PG00D; 37 = USB sense (FT pad)
 38..47 reserved (NC); battery ADC deferred.
 PIO timing/programs NOT validated.

1S Li-ion Battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION
FarmMesh Node
Sheet: /RP2350B aggregation, memory and debug
File: 02-rp2350.kicad_sch

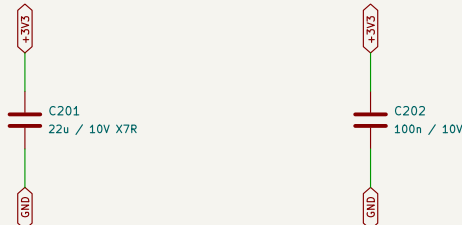
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Size: A3	Date: 2026-09-23
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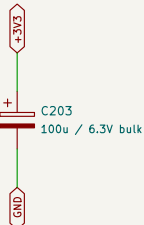
Size: A3	Date: 2026-09-23	Rev: A0 DRAFT
KiCad E.D.A. 10.0.5		Id: 3/7

ESP32-C5 MODULE A / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

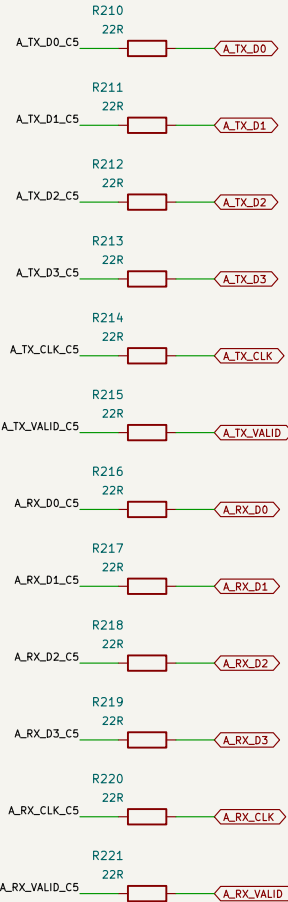
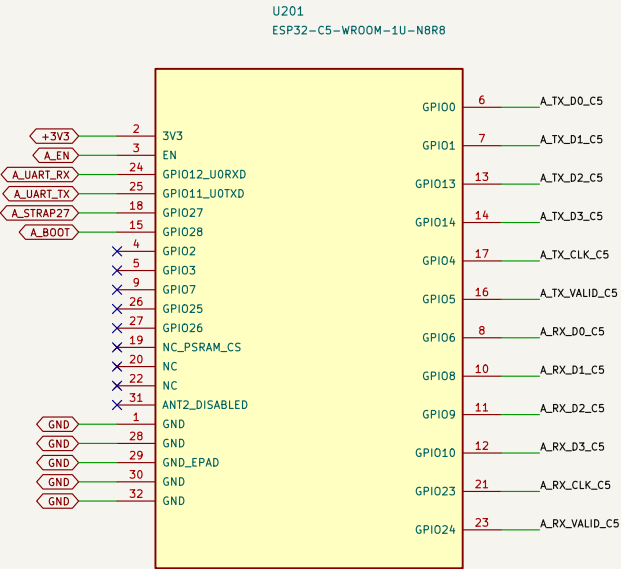
C5 TX and RX clocks both originate here. RP never drives either clock. UART0 download retained; C5 native USB disabled to release GPIO13/14. All interface GPIOs avoid boot straps and internal Flash/PSRAM.



22R initial series damping; validate edges on hardware.



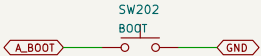
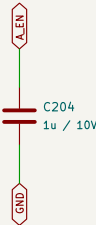
Place 22u + 100n at pad2, short ground return. 100u local reservoir provisional; check full rail stability. ANT1 needs an external 2.4/5 GHz antenna and cable. Pad31 ANT2 remains unused on the default module.



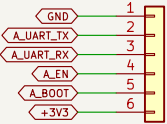
C5 source
C5 source
C5 source
C5 source
C5 source
C5 source
RP source
RP source
RP source
RP source
C5 source
RP source



Place each VALID pull-down at its receiver.



J201
UART / 3V3 reference only



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high. 3.3V logic UART only; header pin6 is a voltage reference, not a power input. USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions. Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION
FarmMesh Node
Sheet: /C5-A module and independent link A/
File: 03-radio-a.kicad_sch

Title: C5-A module and independent link A

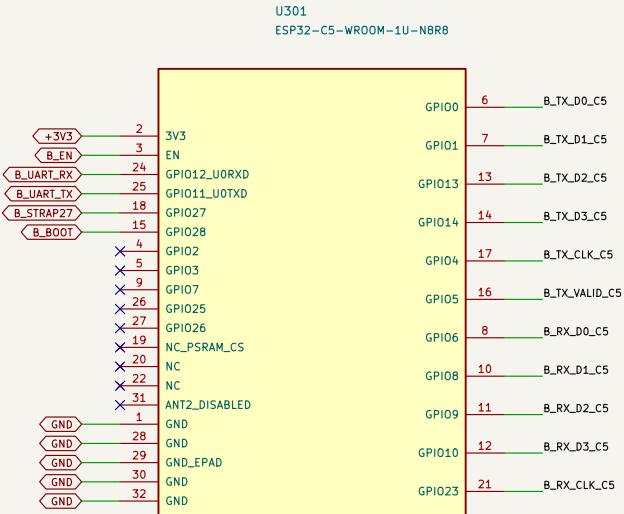
Size: A3 Date: 2026-09-23 Rev: A0 DRAFT
KiCad E.D.A. 10.0.5 Id: 4/7

ESP32-C5 MODULE B / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

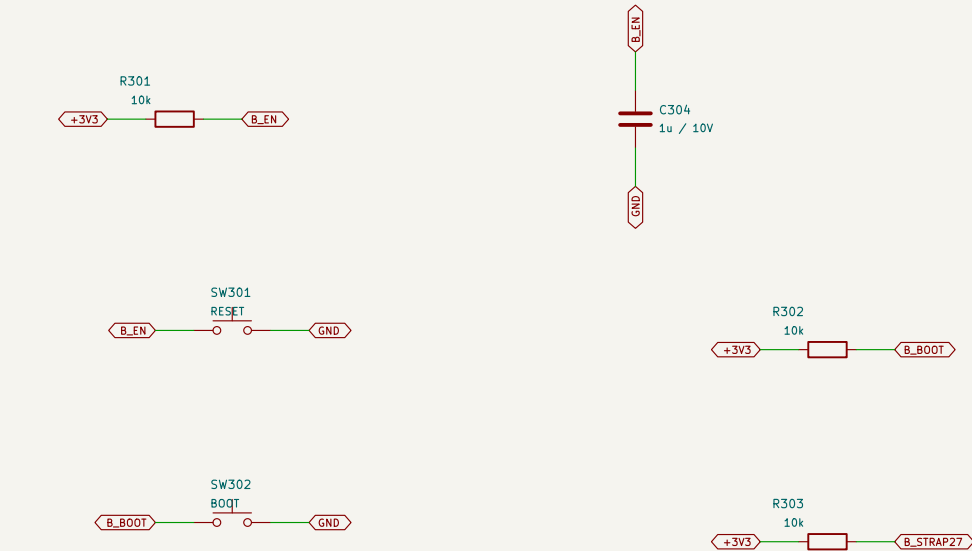
C5 TX and RX clocks both originate here. RP never drives either clock.
UART0 download retained; C5 native USB disabled to release GPIO13/14.
All interface GPIOs avoid boot straps and internal Flash/PSRAM.



Place 22u + 100n at pad2, short ground return.
100u local reservoir provisional; check full rail stability.
ANT1 needs an external 2.4/5 GHz antenna and cable.
Pad31 ANT2 remains unused on the default module.



Place each VALID pull-down at its receiver.



J301
UART / 3V3 reference only



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high.
3.3V logic UART only; header pin6 is a voltage reference, not a power input.
USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions.
Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION

FarmMesh Node

Sheet: /C5-B module and independent link B/
File: 04-radio-b.kicad_sch

Title: C5-B module and independent link B

Size: A3 Date: 2026-09-23

Rev: A0 DRAFT

KiCad E.D.A. 10.0.5

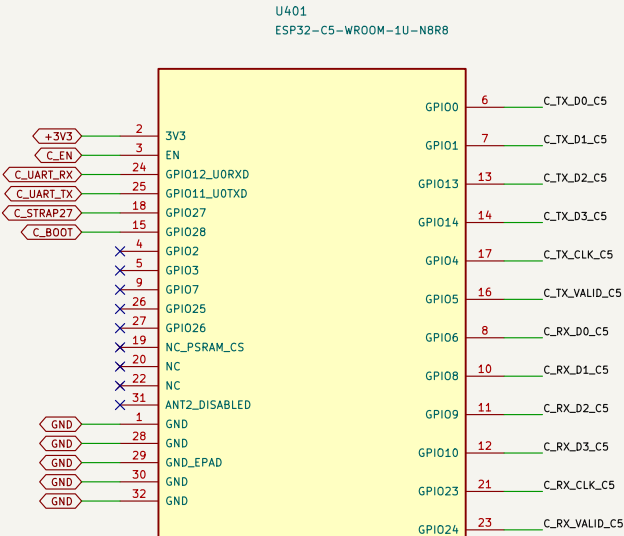
Id: 5/7

ESP32-C5 MODULE C / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

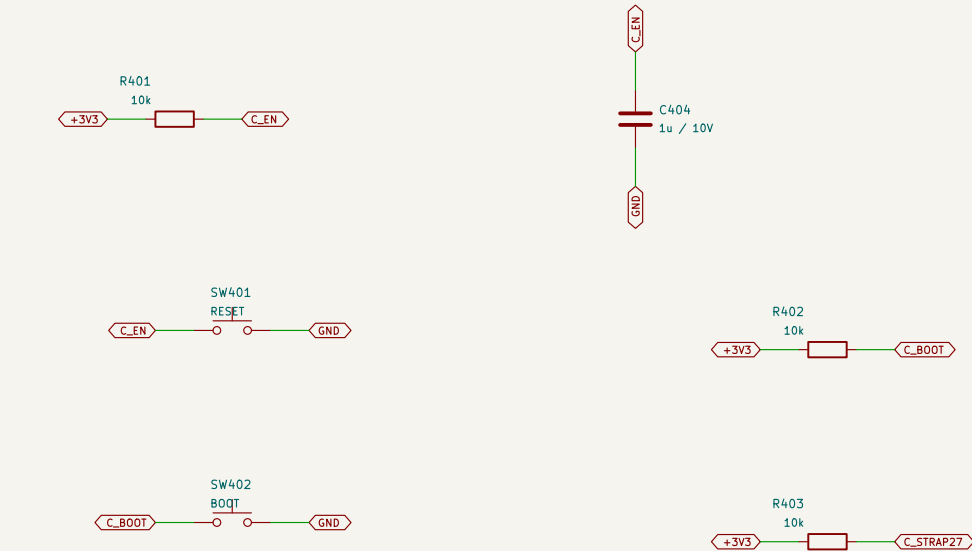
C5 TX and RX clocks both originate here. RP never drives either clock. UART0 download retained; C5 native USB disabled to release GPIO13/14. All interface GPIOs avoid boot straps and internal Flash/PSRAM.



Place 22u + 100n at pad2, short ground return. 100u local reservoir provisional; check full rail stability. ANT1 needs an external 2.4/5 GHz antenna and cable. Pad31 ANT2 remains unused on the default module.



Place each VALID pull-down at its receiver.



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high. 3.3V logic UART only; header pin6 is a voltage reference, not a power input. USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions. Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION

FarmMesh Node

Sheet: /C5-C module and independent link C/
File: 05-radio-c.kicad_sch

Title: C5-C module and independent link C

Size: A3 Date: 2026-09-23

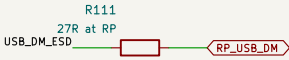
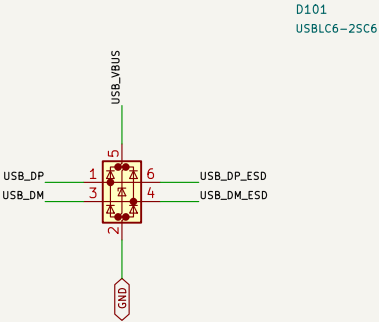
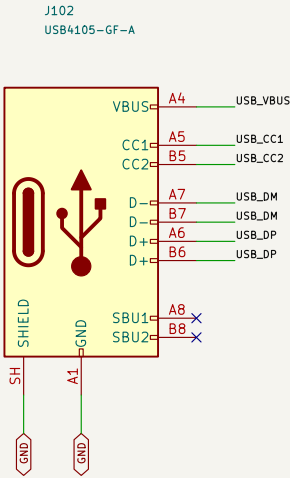
Rev: A0 DRAFT

KiCad E.D.A. 10.0.5

Id: 6/7

RP USB-C / BATTERY POWER REQUIRED

USB VBUS is sensed only. It does not power or charge the battery.
Firmware must gate USB attach on VBUS detection; SWD remains available.
C5 USB pads are allocated to PARLIO; each C5 uses its UART header.



GPI037 is fail-tolerant: divider 5.1k/7.5k gives 2.60–3.30V at VBUS 4.4–5.5V (1%).
Disable GPI037 internal pulls in firmware before testing VBUS.
USB differential route: nominal 90 ohm pair, short ESD path,
27R series parts close to RP. PCB geometry is not designed yet.
No USB-to-battery path, no USB-powered standalone operating mode.

1S Li-ion battery input; no on-board charger
Engineering review only – NOT FOR FABRICATION
FarmMesh Node
Sheet: /RP USB boot / self-powered USB-C/
File: 06-usb.kicad_sch

Title: RP USB boot / self-powered USB-C

Size: A3	Date: 2026-09-23	Rev: A0 DRAFT
KiCad E.D.A. 10.0.5		Id: 7/7